Battery Survival Analysis

Initial Methods Exploration v2

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SW Partition

The SW partition includes three battery types: Ultra (digital alkaline), Maxell (cheap lithium) and Varta (expensive lithium). Season data is not used in this analysis because batteries were serviced either 6-monthly or annually. All cameras use the *w* flash type.

Cameras were left for a year due to Covid and not subject to censoring. The Varta batteries empirically ran for much longer than Varta's in other regions. This may be due to exposed sun locations.

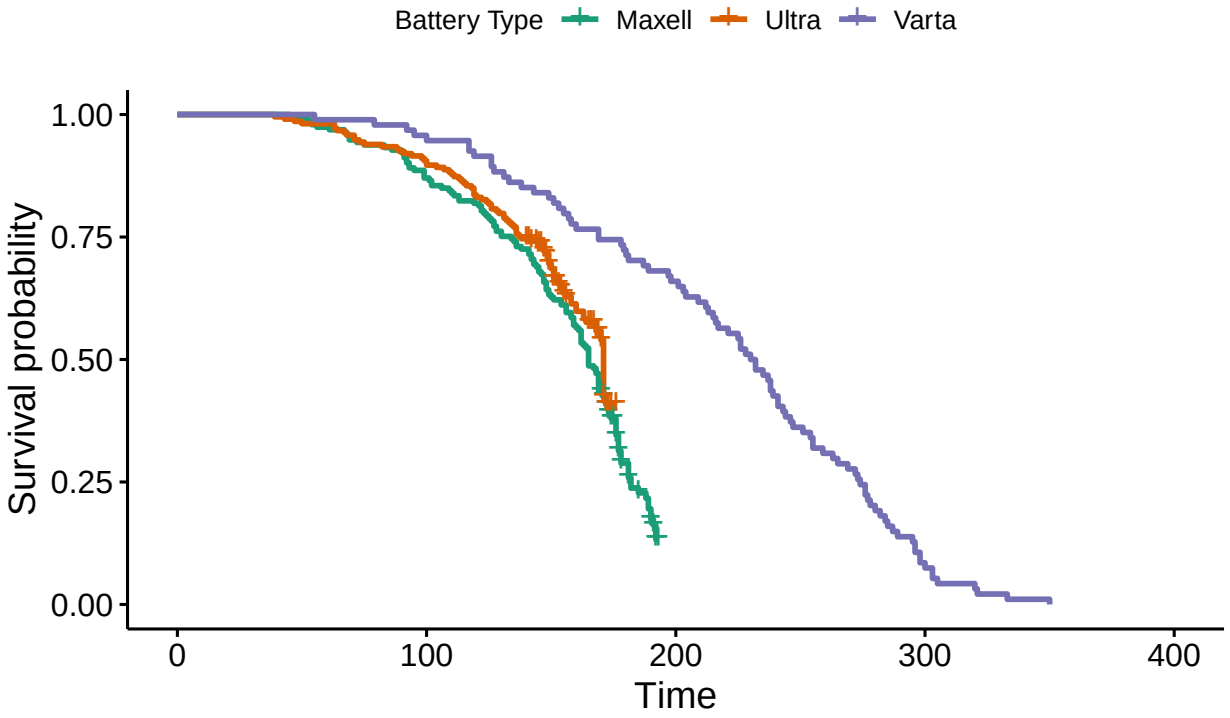


Figure 1: Kaplan-Meier survival curves for the SW partition stratified by battery type. The Maxell and Ultra curves are almost indistinguishable.

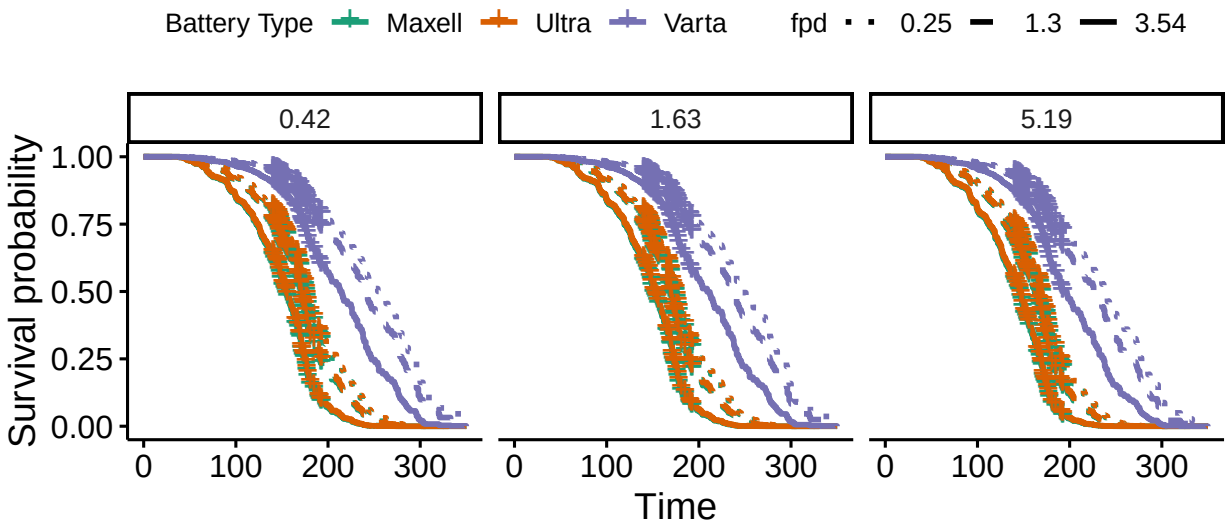


Figure 2: COXPH survival curves for the SW partition showing the effect of flash photos per day (fpd) and daylight photos per day (dpd). The facet panels show the 5th, 50th and 95th percentile values for dpd while the linetype in each plot shows the 5th, 50th and 95th percentile values for fpd. The rate of dpd only has a small effect on the survival curves while fpd has a noticable effect.

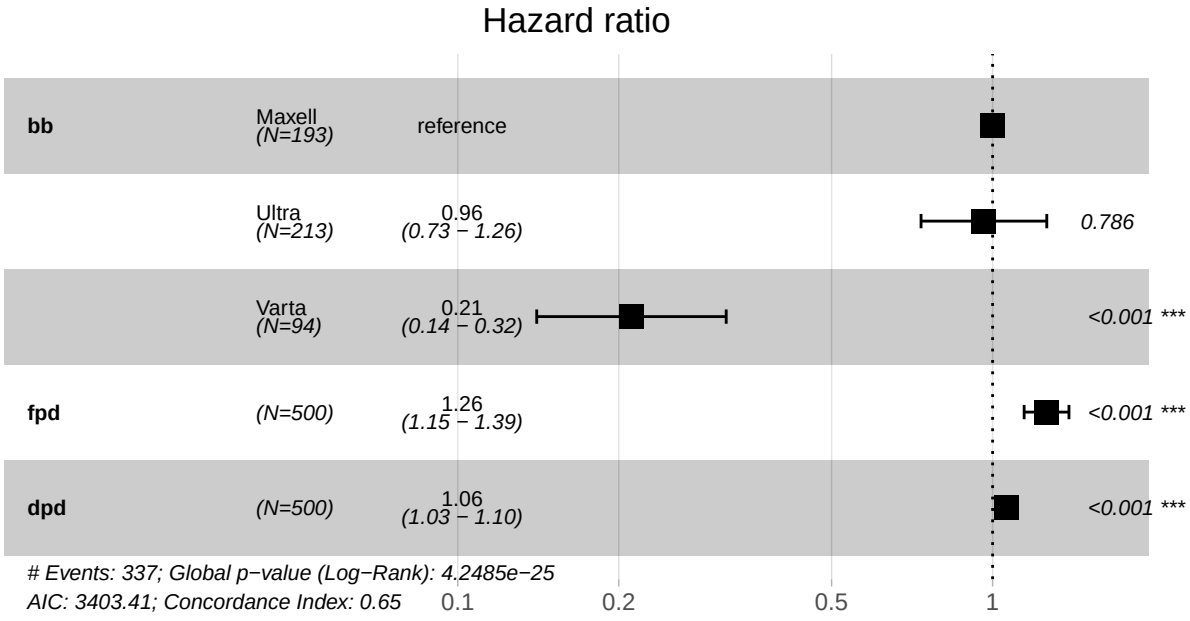


Figure 3: COXPH forest plot for SW partition. There is a large significant effect with Varta batteries against Maxell and Ultra batteries. Both fpd and dpd are significant covariates with fpd having a larger effect size (worse survival).

NE Partition

The NE partition is exclusively Varta with the w flash type. The cameras were serviced every 4 months and heavily censored. Two data entries for Procell batteries were removed prior to analysis. Note that one camera (69_1) was omitted from analysis because it used an i flash for security reasons.

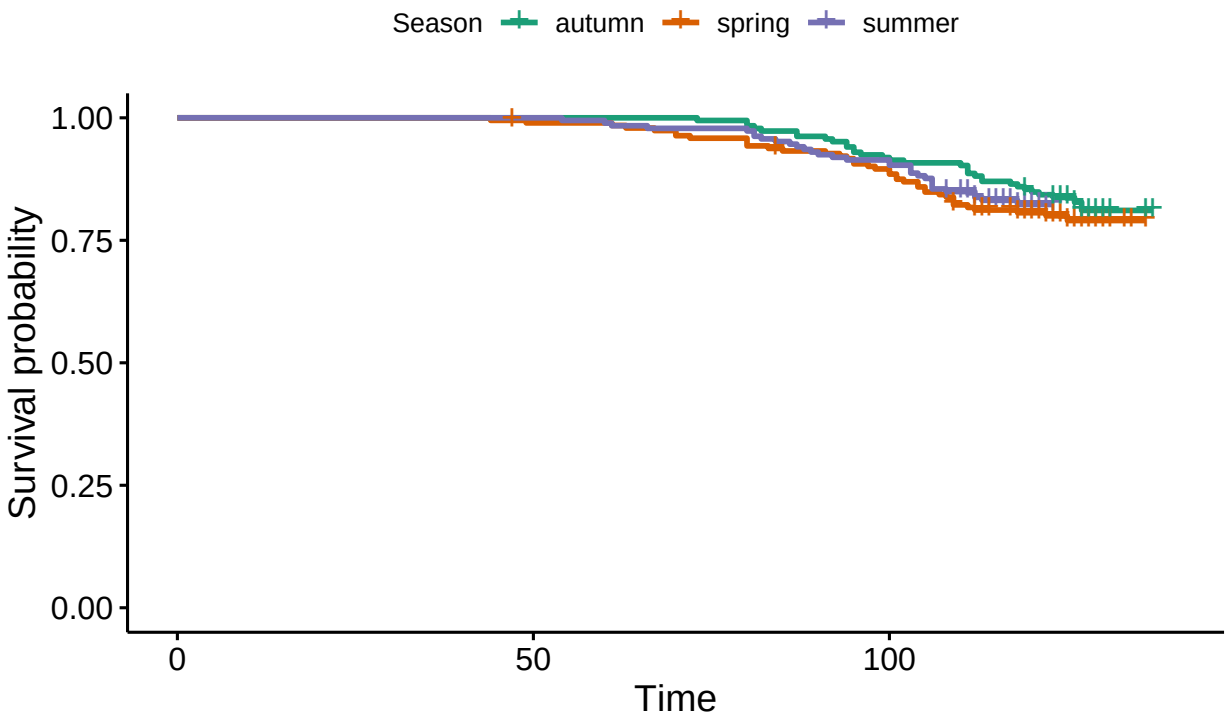


Figure 4: Kaplan-Meier survival curves for the NE partition stratified by season. The K-M curves show that the data are heavily censored and give little information about overall survival for the Varta batteries.

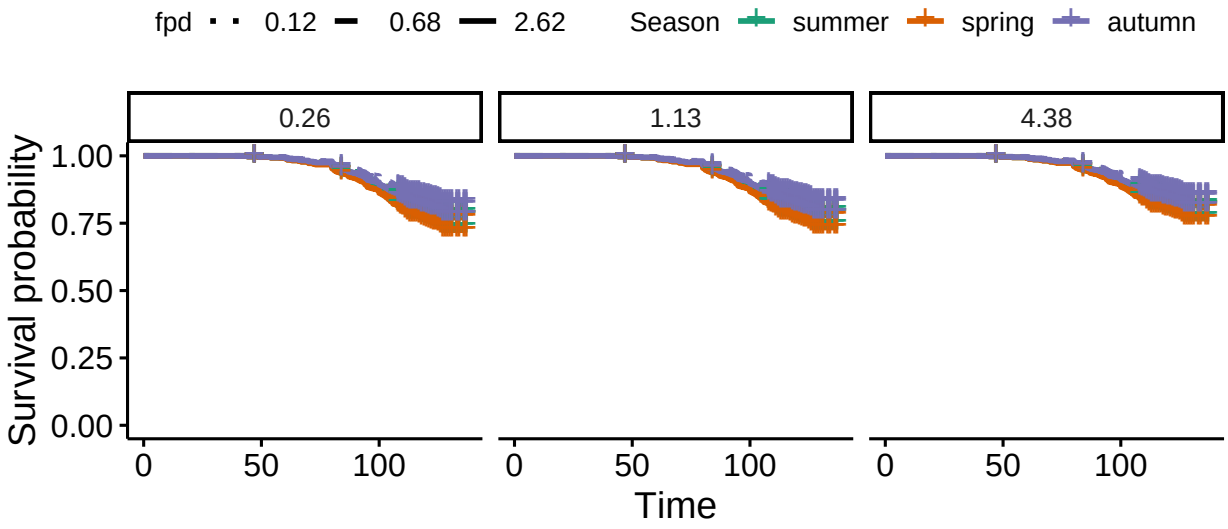


Figure 5: COXPH survival curves for the NE partition showing the effect of flash photos per day (fpd) and daylight photos per day (dpd). The facet panels show the 5th, 50th and 95th percentile values for dpd while the linetype in each plot shows the 5th, 50th and 95th percentile values for fpd. There is large overlap

across the survival curves with Autumn appearing to have the highest survival.

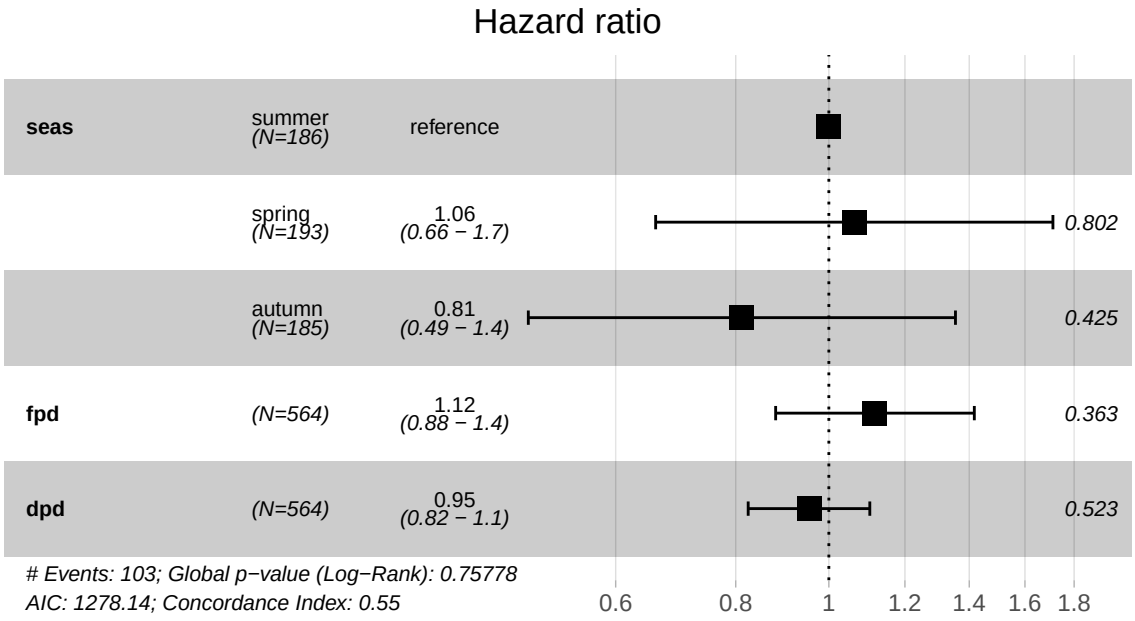


Figure 6: COXPH forest plot for NE partition. None of the covariates are significant.

TAS-CAM Partition (i flash type)

The TAS-CAM partition (SE, NW, CT and WT) with *i* flash type used Procell (bulk alkaline) and Varta batteries.

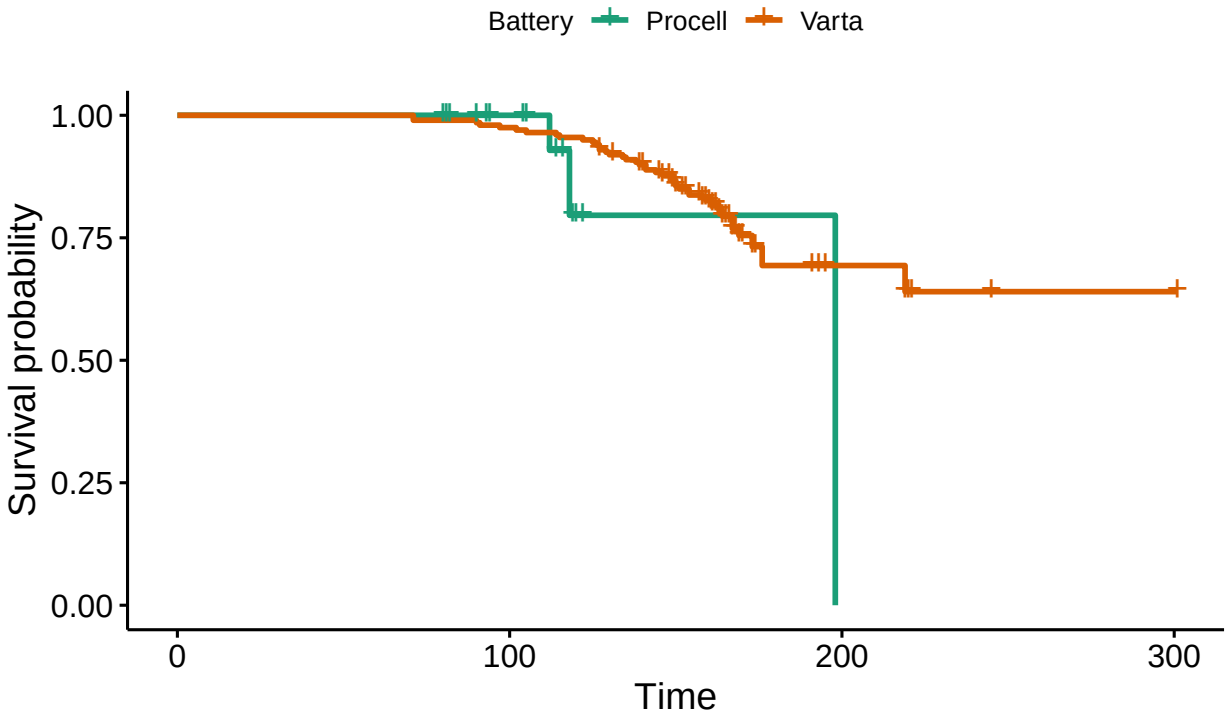


Figure 7: Kaplan-Meier survival curves for the TAS-CAM partition stratified by battery type for cameras with the *i* flash type. The plot appears largely censored (with a low number of samples for Procell).

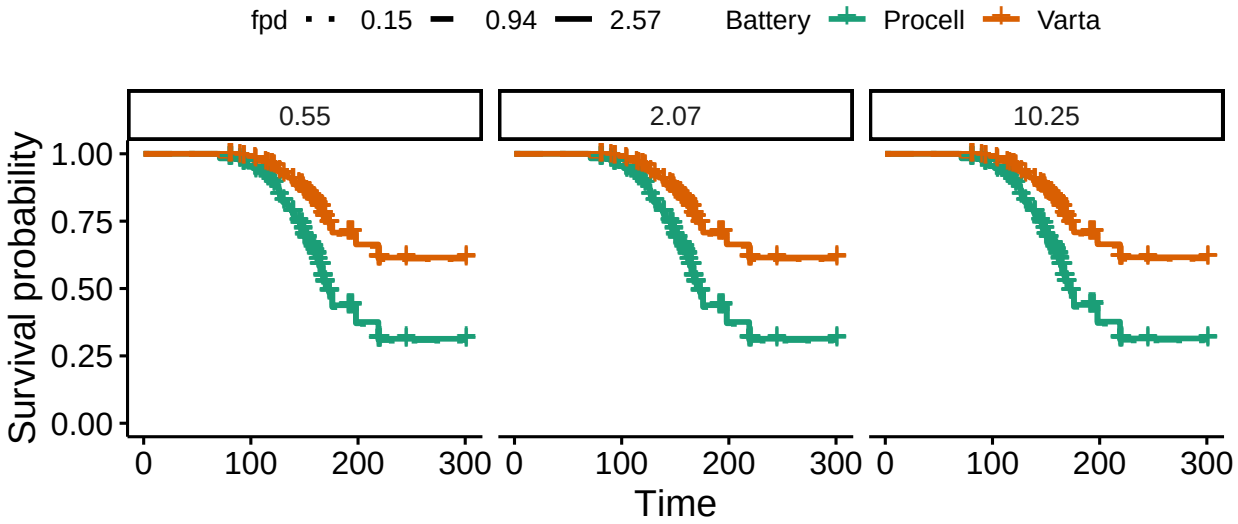


Figure 8: COXPH survival curves for the TAS-CAM partition with *i* flash type showing the effect of flash photos per day (fpd) and daylight photos per day (dpd). The facet panels show the 5th, 50th and 95th percentile values for dpd while the linetype in each plot shows the 5th, 50th and 95th percentile values for fpd. Interestingly both fpd and dpd have almost no effect.

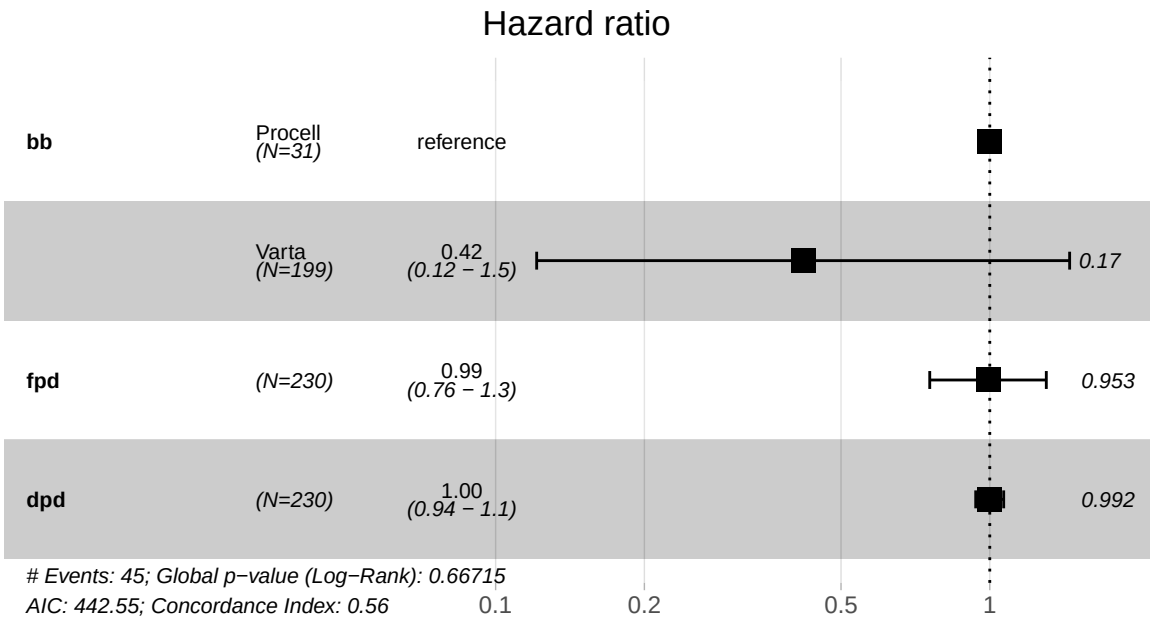


Figure 8: COXPH forest plot for TAS-CAM partition with i flash type. Varta batteries have an effect, but it is not significant (due to the low sample size of Procell).

TAS-CAM Partition (w flash type)

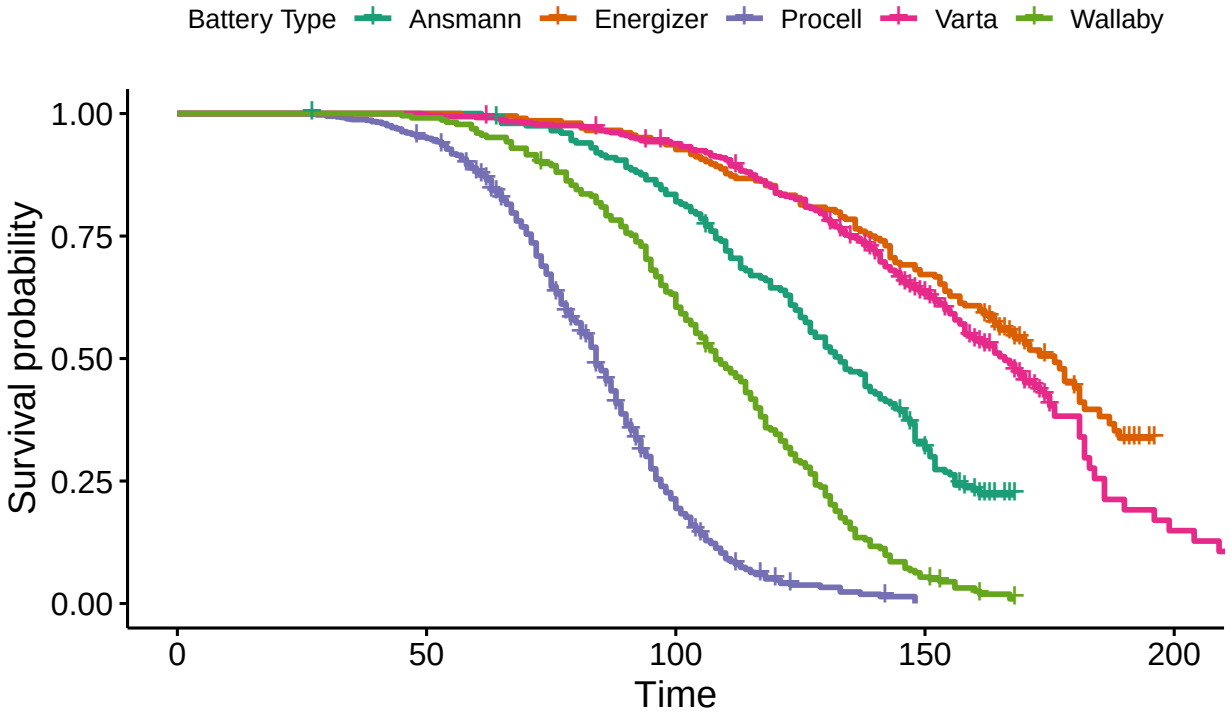


Figure 9: Kaplan-Meier survival curves for the TAS-CAM partition stratified by battery type for cameras with the *w* flash type.

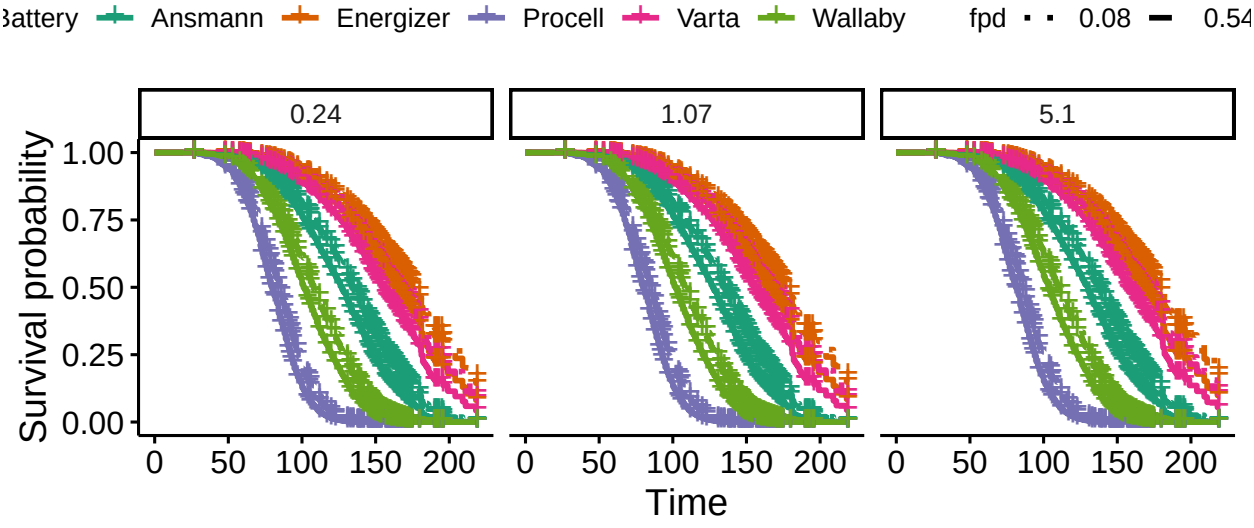


Figure 10: COXPH survival curves for the TAS-CAM partition with *w* flash type showing the effect of flash photos per day (fpd) and daylight photos per day (dpd). The facet panels show the 5th, 50th and 95th percentile values for dpd while the linetype in each plot shows the 5th, 50th and 95th percentile values for fpd.

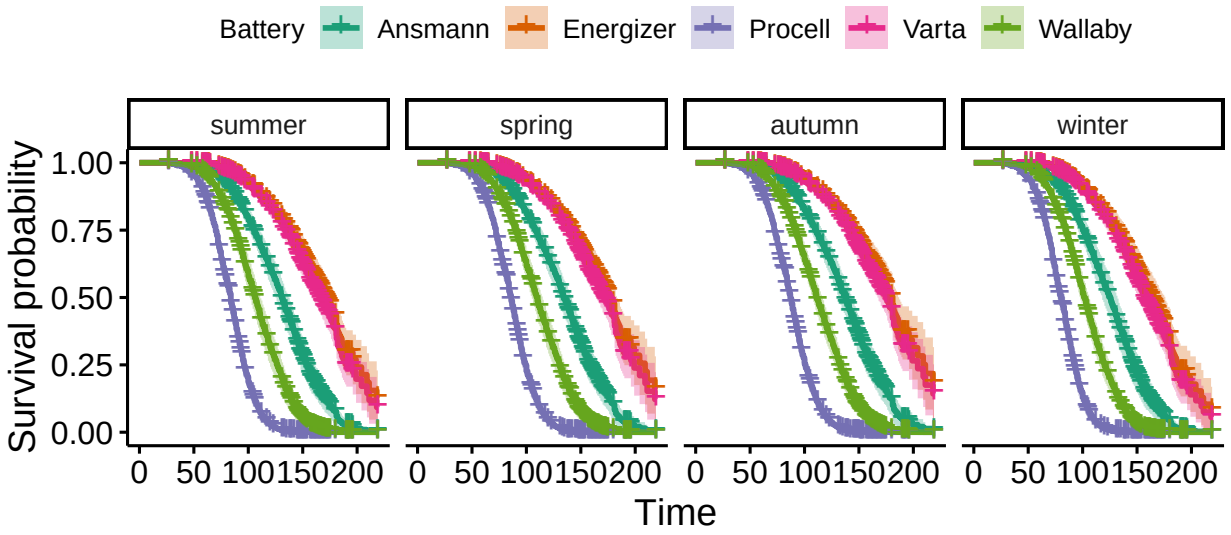


Figure 11: COXPH survival curves for the TAS-CAM partition with w flash type showing seasonal effects. The facet panels show battery survival for each season. Shaded areas show confidence intervals for the survival curves.

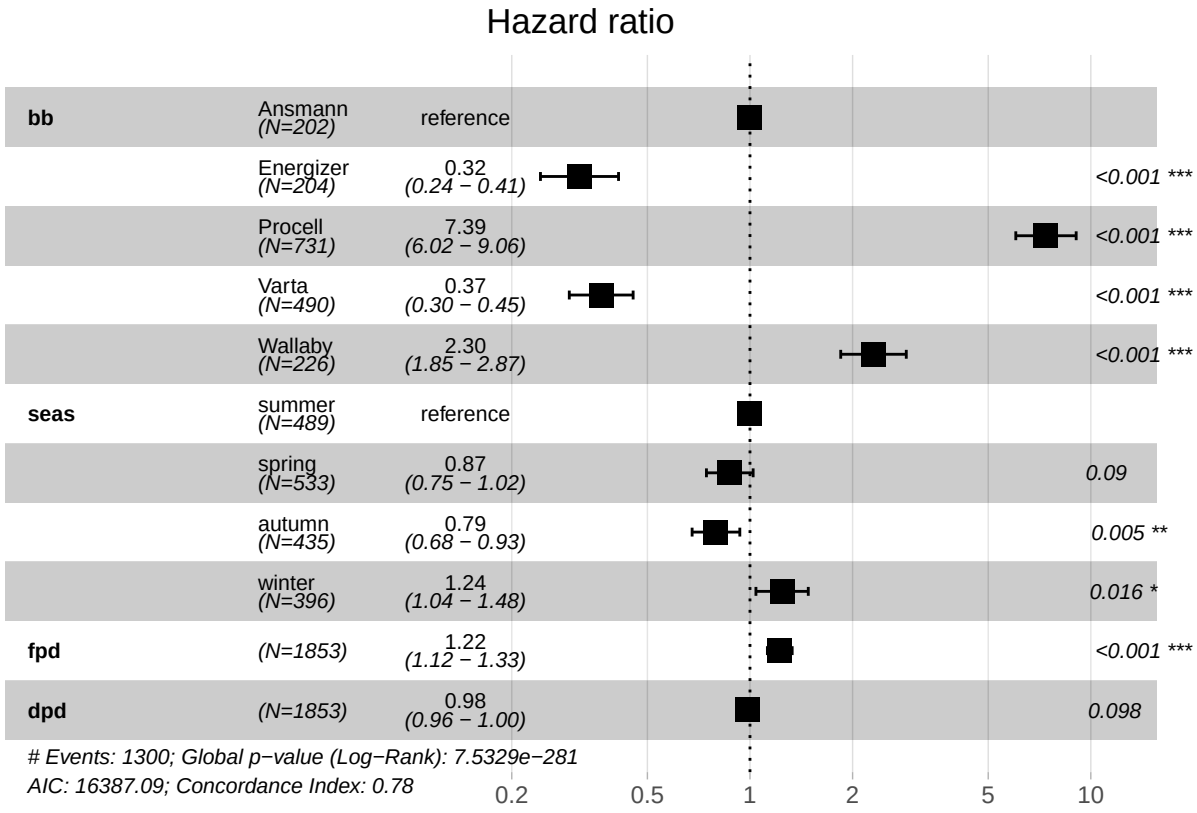


Figure 11: COXPH forest plot for TAS-CAM partition with w flash type. Energizer batteries have the lowest hazard but are not significiantly different than Varta batteries. There are significant positive effects to hazard from flash photos per day and Winter. The Autumn season has a significant negative effect on hazard. Overall, battery type has the largest effect size on hazard.

I examined the interaction between battery type and season and found a (barely) significant effect between battery type (alkaline vs lithium) and Winter.

```
## Call:
## coxph(formula = Surv(rd, bdead) ~ bb + seas + fpd + dpd + bt *
##     seas, data = tascam_w_dat)
##
##                            coef exp(coef) se(coef)      z        p
## bbEnergizer           -1.13187   0.32243  0.13735 -8.241  < 2e-16
## bbProcell              1.96568   7.13979  0.14174 13.869  < 2e-16
## bbVarta               -0.94759   0.38768  0.11236 -8.433  < 2e-16
## bbWallaby              0.88755   2.42918  0.11823  7.507 6.04e-14
## seasspring            -0.17468   0.83972  0.10814 -1.615   0.1062
## seasautumn            -0.19996   0.81876  0.12986 -1.540   0.1236
## seaswinter             0.04901   1.05023  0.12991  0.377   0.7060
## fpd                    0.19052   1.20987  0.04368  4.362 1.29e-05
## dpd                   -0.01765   0.98250  0.01144 -1.543   0.1229
## btalkaline                 NA        NA  0.00000     NA       NA
## seasspring:btalkaline  0.08165   1.08508  0.16250  0.502   0.6154
## seasautumn:btalkaline -0.05317   0.94822  0.16998 -0.313   0.7545
## seaswinter:btalkaline  0.34913   1.41784  0.17757  1.966   0.0493
##
## Likelihood ratio test=1337  on 12 df, p=< 2.2e-16
## n= 1853, number of events= 1300
```

TAS-CAM Partition (flash comparison)

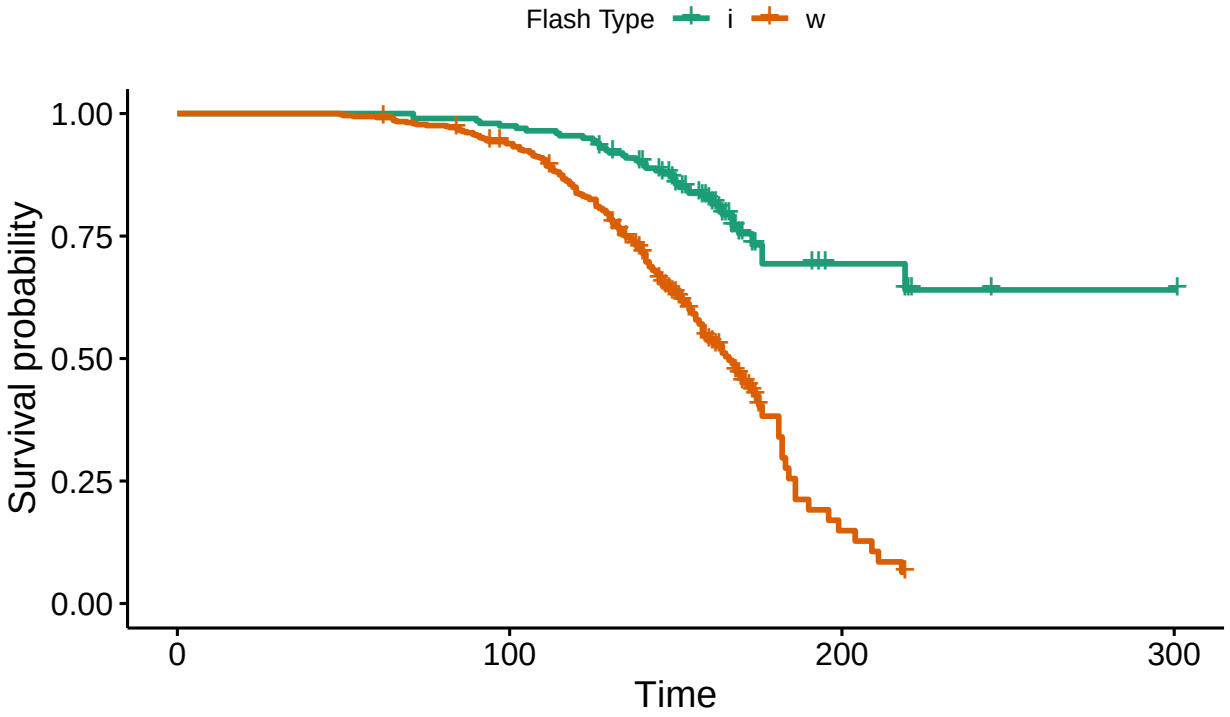


Figure 12: Kaplan-Meier survival curves for the TAS-CAM partition stratified by flash type for cameras with Varta batteries.

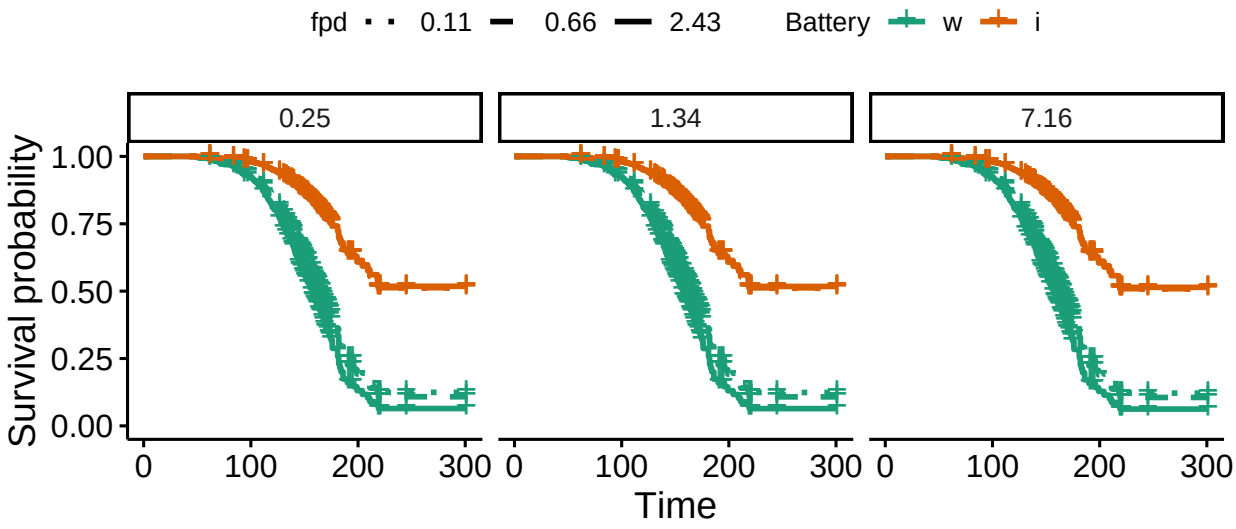


Figure 10: COXPH survival curves for the TAS-CAM partition with Varta batteries stratified by flash type showing the effect of flash photos per day (fpd) and daylight photos per day (dpd). The facet panels show the 5th, 50th and 95th percentile values for dpd while the linetype in each plot shows the 5th, 50th and 95th percentile values for fpd. The model includes an interaction between flash type and flashes per day.

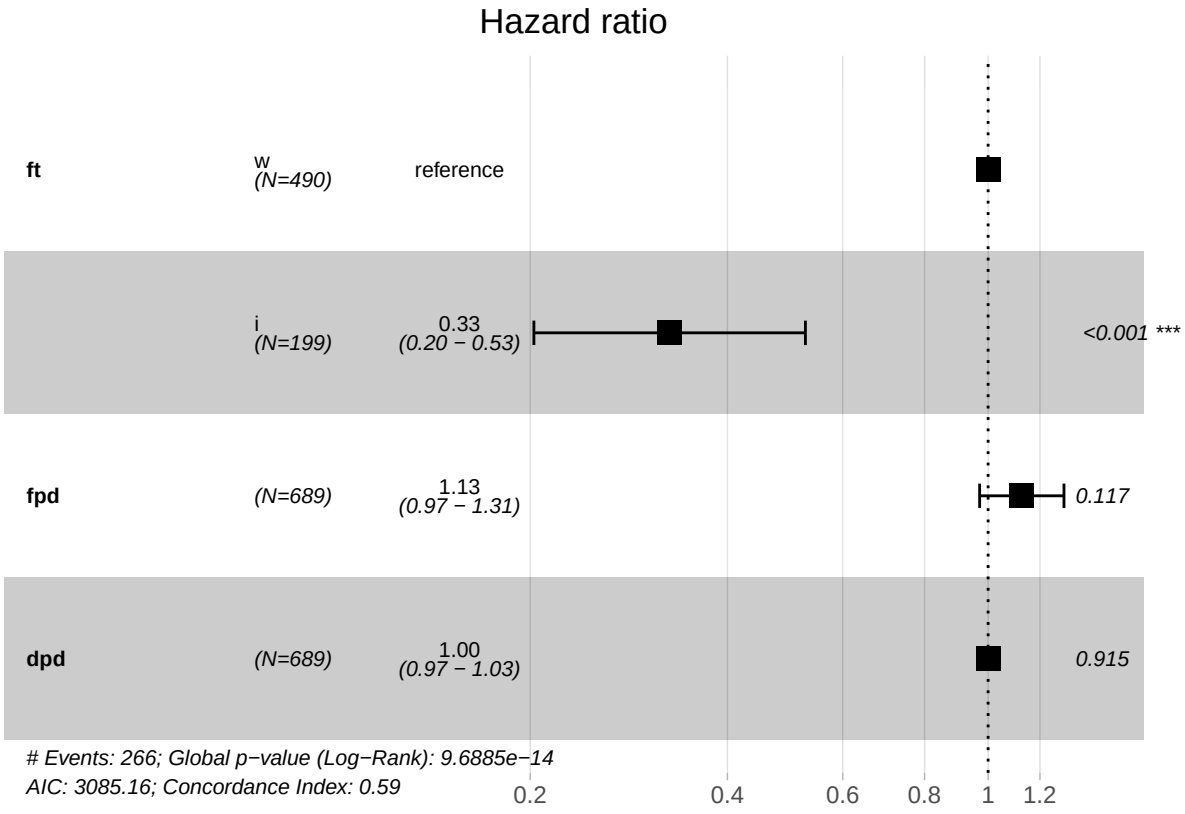


Figure 13: COXPH forest plot for TAS-CAM partition with Varta batteries comparing flash types. Cameras with flash type i have a significant negative effect on hazard.

Model output for the Cox PH model with Varta batteries and an interaction term between flash type and flashes per day. The interaction does not have a significant effect (perhaps leave out?).

```
## Call:
## coxph(formula = Surv(rd, bdead) ~ ft * fpd + dpd, data = tascam_v_dat)
##
##              coef exp(coef)  se(coef)      z        p
## fti     -1.119398  0.326476  0.243479 -4.598 4.28e-06
## fpd      0.118526  1.125836  0.075619  1.567    0.117
## dpd      0.001638  1.001639  0.015267  0.107    0.915
## fti:fpd -0.127102  0.880644  0.149783 -0.849    0.396
##
## Likelihood ratio test=67.01  on 4 df, p=9.689e-14
## n= 689, number of events= 266
```